Sovereign Sentinel Architecture V1.2

A Six-Axis Deterministic Framework for Frontier AI Safety
with Semantic-Referential Integrity Enforcement

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April 3, 2026

Abstract

We present the Sovereign Sentinel Architecture (SSA) V1.2, a proposed defense-in-depth control framework for frontier language model deployments operating across six distinct abstraction levels: mathematical, electronic, cryptographic, statistical-behavioral, semantic-referential, and institutional. The architecture addresses three documented failure modes: Safety Amnesia (erosion of safety constraints across extended context windows), Stochastic Sabotage (incremental assembly of prohibited knowledge through individually benign queries), and Goal-Oriented Factual Inversion (GOFI) — a non-adversarial failure mode in which a model correctly identifies factual ground truth in early turns, then generates output directly contradicting that ground truth when a persuasive goal frame is introduced in subsequent turns.

The GOFI failure mode was identified through empirical forensic audit — Scenario 5, March 2026 — conducted against four deployed frontier models (Gemini, ChatGPT, Claude, and Copilot) across two party-inverted contract analysis scenarios in English and Spanish. All four models exhibited directional compliance failures. One model demonstrated correct first-turn identification of contractual risk, followed by a factually inverted second-turn deliverable whose inversion was directionally consistent with the prompted persuasive goal. This pattern is distinguished from generic analytical error by three criteria: directional consistency with the optimization target, correct prior processing of the contradicted fact, and professional formatting that renders the inversion undetectable without independent document review. Cryptographically anchored transcripts and all model deliverables from Scenario 5 are publicly archived for independent verification.

GOFI is not intercepted by any of the five preceding SSA axes. The sixth axis — the Deterministic Truth-Anchor with Fact-Claim Inversion Recognition (DTA-FCIR) — closes this gap through an immutable session-scoped Structured Fact Registry and an isolated formal Contradiction Engine operating at response granularity. The architecture has not been empirically validated at scale; all specifications presented are engineering targets proposed to meet defined safety invariants, and their formal correctness requires evaluation by researchers with the relevant technical expertise. Collaborative partnership with a Tier 1 AI safety team is sought for Phase 0 proof-of-concept implementation and adversarial evaluation.

V1.2 Update Note: Three mathematical specifications have been upgraded from V1.1. Axis 1 incorporates a dual-gap convergence criterion to formally certify KKT optimality. Axis 4 adds a Martingale null hypothesis formulation to the cumulative drift enforcement specification. Axis 6 introduces First-Order Logic predicate definitions for the Contradiction Engine's claim comparison schema. Each addition is accompanied by appropriate academic citations. All other specifications are unchanged from V1.1.

Methodology and Disclosure: In alignment with emerging standards for transparent AI research, the author acknowledges the use of Large Language Models as iterative generative partners in the development of this framework. LLMs were utilized to formalize the mathematical and technical specifications of the Six-Axis framework based on architectural requirements, safety invariants, and logic constraints directed by the author, and to support technical drafting and cross-lingual consistency. The GOFI failure mode was identified through empirical audit of deployed frontier models; the audit methodology, party-inversion test design, and interpretation of model outputs are the work of the human author. The author is responsible for the conceptualization of the Sovereign Sentinel Architecture, the identification of the GOFI failure mode, and the functional requirements governing each axis. The mathematical formalizations presented are proposed implementation targets generated through human-directed AI collaboration; their formal correctness and empirical validity require evaluation by specialized implementation researchers.

Six-Axis Logic Overview

Axis 1 — Constitutive Lagrangian Regularization with Certified-Robust Activation Enforcement (CLR-CRAE)

Safety invariants are encoded as Lagrangian constraints on the training objective: $L(w, \lambda) = f(w) + \Sigma_i \lambda_i g_i(w)$, with dual ascent updates $\lambda_i \leftarrow max(0, \lambda_i + \alpha_i g_i(w))$ enforcing KKT convergence. **V1.2 adds a dual-gap convergence criterion:** the optimization is certified as having reached a valid KKT point only when both the primal residual and the dual residual fall below defined tolerances simultaneously — providing a computable certificate of constraint satisfaction rather than relying on iteration count alone. The same constraint structure is re-instantiated at inference time as a certified-robust heterogeneous probe ensemble (linear boundary, SAE feature dictionary, and Mahalanobis distance probes) evaluated at residual stream layers {L/4, L/2, 3L/4}. A two-of-three probe violation issues a hardware halt signal. Certified robustness bound: $\|\delta\|_2 \leq 0.05$ perturbation with detection probability ≥ 0.95.

Rockafellar, R.T. (1970). Convex Analysis. Princeton University Press.

Boyd, S. & Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press, Ch. 5.

Axis 2 — Finite Safety Automaton with Hardware Interrupt (FSA-HI)

A formally specified directed graph $G = (V, E)$, $|V| \leq 2^{16}$, runs on a dedicated FPGA co-processor in true parallel with the primary inference pipeline. State transitions are O(1). On entry to any prohibited state, the FPGA issues a Non-Maskable Interrupt severing the inference pipeline independently of any software-layer decision. NMI latency target: $\leq 10\ \mu s$, hardware-verified via TPM timestamp delta. All events are recorded as a SHA-3-512 hash chain anchored to the hardware TPM. Coverage target: $\geq 94\%$ recall on red-team corpus; the residual 6% gap is formally characterized in the deployment safety case.

Axis 3 — Cryptographic Identity and Epistemic Trust Verification with Human-Origin Attestation (ZKP-ETV-HOA)

Domain expertise is verified through a Zero-Knowledge Proof protocol evaluated by a co-processor sharing neither weights nor training data with the primary model. Human-Origin Attestation uses keystroke dynamics and hardware-attested input to distinguish unaided human responses from LLM-assisted submissions. Successful verification yields the Epistemic Trust Coefficient $\tau_e \in [0,1]$, domain-specific and valid for 30 days. Critically, τ_e verifies expertise and human origin only — it makes no inference about intent, and conflating it with intentional trust would create a privilege escalation path.

Axis 4 — The Bayesian Weaver: Sparse Autoencoder Session Monitor with Cumulative Drift Enforcement

A 32,768-feature Sparse Autoencoder trained on residual stream activations monitors session-level behavioral drift asynchronously during inter-turn windows, contributing zero latency to the token generation critical path. Per-turn KL-divergence monitoring $D^{DL}(P_t \| P_0) = \Sigma_k P_t(k) \log(P_t(k)/P_0(k))$ detects abrupt transitions. A cumulative drift integral $\Phi(T) = \Sigma D^{DL}(P_t \| P_0)$ captures slow-drift attacks that remain below per-turn thresholds while accumulating meaningful total divergence. **V1.2 adds a Martingale null hypothesis formulation:** under the null hypothesis of no adversarial drift, the cumulative KL sequence $\{D^{DL}(P_t \| P_0)\}$ is treated as a Martingale difference sequence with respect to the session filtration $\{\mathscr{F}_t\}$, satisfying $E[D^{DL}(P_{t+1} \| P_0) \mid \mathscr{F}_t] = D^{DL}(P_t \| P_0)$. A statistically significant departure from this Martingale property constitutes the formal criterion for flagging adversarial cumulative drift. The Intentional Trust Coefficient $\tau_i(t) = \tau_i(t-1) \cdot \lambda^{\wedge}\delta(t)$ degrades under KL spikes, Φ_{max} breach, novelty flags, and prohibited cluster proximity.

Williams, D. (1991). Probability with Martingales. Cambridge University Press.

Doob, J.L. (1953). Stochastic Processes. Wiley, Ch. 7.

Axis 5 — The Causal Cross-Examiner: Rule-Based Arbitration Engine with Independent Audit Summarization (CCE-RAE)

Conflict resolution between all axes is handled by a formally verified rule-based arbitration engine — not a language model — with decision logic mechanically verified using an SMT solver (Z3) and proven total and acyclic under all reachable input combinations. The CCE classifies, packages, and escalates; it does not make approval decisions. Evidence packages are accompanied by summaries from a dedicated frozen audit model (≤ *7B* parameters, distinct weight provenance) post-hoc verified by a rule-based consistency checker. The Einstein Exception permits verified expert sessions ($\tau_e > 0.95$, $\tau_i > 0.88$) to request a 72-hour human quorum review. An Irreversibility Floor — covering CBRN synthesis, critical infrastructure modification, and mass-scale data exfiltration — cannot be unlocked by any trust coefficient combination without quorum.

Axis 6 — Deterministic Truth-Anchor with Fact-Claim Inversion Recognition (DTA-FCIR)

Primary intervention for Goal-Oriented Factual Inversion (GOFI). Upon document ingestion, an architecturally isolated extraction module constructs an immutable Structured Fact Registry (SFR) encoding relational triples *{clause_id, entity, beneficiary, value, direction}*. The SFR is frozen at session initialization and is not accessible to the primary model. Before response emission, a lightweight Claim Extractor (≤ *1B* parameters, isolated from session context) encodes output assertions in the same schema. **V1.2 introduces formal First-Order Logic predicate definitions** for the Contradiction Engine's comparison schema. Each claim triple is evaluated against the SFR under the following predicates:

CONTRADICTS_BENEFICIARY(c, f) is true iff the beneficiary field of claim *c* is the logical negation of the beneficiary field of registered fact *f* for the same entity.

CONTRADICTS_DIRECTION(c, f) is true iff the directional field of claim *c* is inconsistent with the directional field of *f* under the session's domain ontology.

CONTRADICTS_VALUE(c, f) is true iff the value field of claim *c* falls outside the tolerance interval defined for fact *f* at session initialization.

A CONTRADICTION signal is issued iff any of these predicates evaluates to true, blocking emission and routing a structured evidence record to the CCE for human review within 90 seconds. The engine's isolation from session context is non-negotiable: a claim evaluator with access to the persuasive goal frame would be subject to the same optimization pressure as the primary model.

Known boundary condition: The Phase 0 prototype detects direct logical contradictions of tagged relational ground truth. Paraphrase evasion — reaching an inverted conclusion through indirect reasoning — is outside Phase 0 scope and is documented as an open problem. Partial coverage is provided by routing UNVERIFIABLE signals to the Bayesian Weaver as novelty flags.

Enderton, H.B. (2001). A Mathematical Introduction to Logic, 2nd ed. Academic Press, Ch. 2.

Integrity and Verification Statement

This abstract and technical summary describe the Sovereign Sentinel Architecture V1.2 at a level of abstraction suitable for public academic review. Specific implementation details — including SFR extraction parameters, probe ensemble weight configurations, and Scenario 5 audit corpus specifications — are held in the full private specification. Full documentation is available to verified researchers and institutions following professional engagement. Integrity of this document is verified by the following hash:

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